# Anthony A. Snead

Department of Biology
New York University
New York, NY 10003
anthony.snead@nyu.edu | anthony-snead.com | @AnthonyASnead

## EDUCATION

**Ph.D. Biology** **Fall 2018 – Spring 2023**

University of Alabama, Tuscaloosa, AL
Dissertation**:** Environmental Responses Across Biological Scales: Abiotic Drivers of Species Range, Population Size, and Genetic Variation in the Mangrove Rivulus, *Kryptolebias Marmoratus*
Committee Members: Ryan L. Earley, Julia Cherry, William Ellis, Paige Ferguson, Jeff Lozier
GPA: 4.0

**B.S. Environmental Biology** **Fall 2016 - Summer 2018**

University of South Florida
GPA: 3.75

## PUBLICATIONS

### *Peer-Reviewed*

X. **Snead AA**, Quackenbush CR, Trojahn S, McDonald A, Lins L, Cornelius C, Adams PE, Ma D, Haag E, Silvestre F, Kanamori A, Earley RL, Kelley JL (202X) Embryonic thermal environments drive plasticity in gene expression. ***Submitted:*** *Proceedings of the Royal Society B: Biological Sciences*

X**. Snead AA**, Marson K, Earley RL (202X). Disentangling Associations between abiotic variables, biotic variables and species abundance in the intertidal. ***Under Revision***: *Journal of Animal Ecology*

6. **Snead AA**, Tatarenkov A, Avise JC, Taylor DS, Turner BJ, Marson K, Earley RL (2023). Out to sea: Ocean currents and patterns of asymmetric gene flow in an intertidal fish species. *Frontiers in Genetics* 14 (doi: 10.3389/fgene.2023.1206543)

5. Clark RD, Fitz KS, Garcia E, Jaynes KE, Reid BN, Sawkins A, **Snead AA**, Whalen J, Pinsky ML (2023) The Practice and Promise of Temporal Genomics for Measuring Evolutionary Responses to Global Change. *Molecular Ecology Resources* (doi: 10.1111/1755-0998.13789).

4. **Snead AA** & Clark RD (2022). The Biological Hierarchy, Time, and Temporal 'Omics in Evolutionary Biology: A Perspective. *Integrative & Comparative Biology* 62(6), 1872-1886 (doi: 10.1093/icb/icac146).

3. **Snead AA** & Alda F (2022). Time Series Sequences for Evolutionary Inferences. *Integrative & Comparative Biology* 62(6), 1771-1783 (doi: 10.1093/icb/icac138).

2. **Snead AA** & Earley RL (2022). Predicting the in-Between: Present and future habitat suitability of an intertidal euryhaline fish. *Ecological Informatics* 68:101523 (doi: 10.1016/j.ecoinf.2021.101523)

1. Davidson J, Summerfelt S, Espmark AMO, Mota V, Marancik D, Earley RL, **Snead AA**, Good C (2021). Effects of ozone on post-smolt Atlantic salmon (*Salmo salar*) performance, health, and maturation in freshwater recirculation aquaculture systems. *Aquaculture* 533:736208 (doi: 10.1016/j.aquaculture.2020.736208)

### *Special Issues*

1. Organizer/Associate Editor, *Integrative & Comparative Biology*, 2022. Special issue on "Temporal Genomics."

### *Media Coverage*

1. Schmelzer H. 2022. Microevolution can guide conservation efforts. *NatureVolve* 11:26-29.
   ** Interview & Popular Science Article Spotlight*

### *General Publications*

1. Snead AA (2023) Networking and Establishing Successful Collaborations. Early Career Ecology Section. https://www.esa.org/earlycareer/2023/06/16/networking-and-establishing-successful-collaborations/

## GRANTS AND FELLOWSHIPS ($372,887.50 total)

### *Accepted*

12. National Science Foundation Postdoctoral Research Fellowship in Biology **$280,000**
*Salamanders in the City: An Integrative Investigation into Microevolutionary Change*

11. Ecological Society of America Conference Funding (Southeast Chapter) **$400**
*Montreal: 2022*

10. University of Alabama Graduate Travel Award **$500**
*Summer 2022*

9. University of Alabama Aquatic Biology Enhancement Assistantship **$54,538**
*2022-2023 Academic Year*

8. RCN for Evolution in Changing Seas 2020 Call for Working Groups **$15,762.50**
*Implications of Methodology on Temporal Genomic Inferences*
(One of Three Project Leads out of 9 Collaborators)

7. RCN for Evolution in Changing Seas **$500**
*2020 Virtual Lab Meeting Training Program*

6. 25th Summer Institute in Statistical Genetics Scholarship **$900**
*Module 9, 14, & 17*

5. E.O. Wilson Biodiversity Fellowship Summer 2020 **$6,944**
*Population Genetic and Environmental DNA Sampling in the Florida Keys*

4. Bishop-Stackman Marine Science Endowed Scholarship **$500**
*Environmental DNA Sampling in the Florida Keys*

3. Office of Research and Economic Development Small Grant Program **$5,844**
*Validating a Non-Invasive Method for Estimating Abundance in Threatened or Cryptic Marine Fishes.*

2. Sigma Xi Grants In Aid of Research **$1,000**
*Escaping the Patch: The Role of Ocean Currents in Determining Population Connectivity in Kryptolebias marmoratus.*

1. The College Academy of Research, Scholarship, and Creative Activity Grant **$5,999**
*If You Can't See It, It Doesn't Exist. Or Does It? Validating a Technique to Estimate Abundance in a Vulnerable, Highly Enigmatic Marine Fish.*

### *Declined*

1. Presidential Management Fellowship (PMF)
*Class of 2023*

## TALKS AND SEMINARS

### *Organized*

Temporal Genomics Lecture Series (Virtual) **2021**
Funded by RCN-ECS

### *Invited*

Stranded within the Intertidal: Species-specific Responses to Environmental Change **2022**
Fresno State (Virtual; April 1st, 2022)

Crossing Disciplines: An Integrative Approach to Evolutionary Ecology **2021**
Saint Leo University (Virtual; February 9th, 2021)

## PRESENTATIONS

### *Oral*

Plastic Gene Expression in Response to Embryonic Thermal Conditions **2023**
**Anthony A. Snead**, Corey R. Quackenbush, Shawn Trojahn, Anna McDonald, Luana Lins, Chris Cornelius, Paula E Adams, Dengke Ma, Yuying Hsu, Eric Haag, Frédéric Silvestre, Akira Kanamori, Ryan L Earley, Joanna L. Kelley
Society of Integrative and Comparative Biology 2023

Temporal Genomics Working Group Update **2022**
René Clark, Brendan Reid, Allyson Salazar Sawkins, John Whalen, Katrina Catalano, Kyra Fitz, Eric Garcia, Kyle Jaynes, **Anthony A. Snead**, Malin Pinsky
RCN for Evolution in Changing Seas 2022 Integration and Training Workshop

Life within the intertidal: Abiotic and biotic drivers of species abundance **2022**
**Anthony A. Snead**, Kristy Marson, Ryan L. Earley
CSEE & ESA 2022 Joint Meeting

Left behind: habitat suitability shifts of a euryhaline fish **2022**
**Anthony A. Snead** & Ryan L. Earley
FSBI Symposium *(3-minute Speed Talk & Poster)*

### *Poster*

Temperature-induced canalization of gene expression in a genetically homogenized population **2023**
**Anthony A. Snead**, Corey R. Quackenbush, Shawn Trojahn, Anna McDonald, Luana Lins, Chris Cornelius, Paula E Adams, Dengke Ma, Yuying Hsu, Eric Haag, Frédéric Silvestre, Akira Kanamori, Ryan L Earley, Joanna L. Kelley
SMBE Satellite Meeting: Small Populations

Validating a non-invasive method for estimating abundance in threatened or cryptic marine fishes **2021**
Ryan L. Earley & **Anthony A. Snead**
University of Alabama Faculty Research Day

## WORKSHOPS ATTENDED

Container Camp 2023 **2023**
**CyVerse**

Spatial Phylogenetics Workshop **2022**
**International Biogeography Society**

The Statistics Wars and Their Casualties **2022**
**Dr. Deborah Mayo, Dr. Roman Frigg, & Dr. Margherita Harris**

Explore & work with harmonized continental-scale biodiversity data from NEON and the US LTER **2022**
**Ecological Society of America (ESA 2022)**

Write a good title, give a better talk **2022**
**Ecological Society of America (Early Career Ecologist Section)**

Coral Health Index Workshop **2022**
**Dr. Michael Sweet, Dr. Raquel Pixoto, & Dr. Christian Voolstra**

Workshop on Molecular Evolution **2022**
**The Marine Biological Laboratory, University of Chicago**

Statistical Methods Seminar Series **2022 - 2023**
*Zero Inflated GLM and GLMM*
*Integrated Step-Selection Analysis*
*Multi-Species (Species Interactions) Occupancy Modeling*
*Nimble*
*Hidden Markov Models in Ecology*
*Integrated Species Distribution Models*
**Ecological Forecasting Initiative & ESA Statistical Ecology Section**

High-Performance Computing Cluster Workshop Series **2021**
*Pytorch*
*Miniconda*
**University of Alabama**

25th Summer Institute in Statistical Genetics- Funded ($900) **2020**
*Module 9: Quantitative Genetics*
*Module 14: Association mapping: GWAS and Sequencing Data*
*Module 17: Computational Pipeline for WGS Data*
**University of Washington**

5th edition Population Genomics **2020**
**Physalia**

Southeastern Computational School: eDNA and QIIME 2 Software Training **2018**
**The University of Tennessee at Chattanooga**

## RESEARCH EXPERIENCE

**Postdoctoral Research Fellow** **Summer 2023 - Present**
*The Winchell Lab, New York University*

- Assembles and annotates large complex genomes.
- Quantifies phenotypic differentiation in 3D morphology, thermal performance, and desiccation risk driven by urbanization in amphibians and insects.
- Uses genomic evidence to identify the consequences of urbanization and climate change in amphibians.
- Mentors and guides graduate and undergraduate students in research projects aimed at understanding the ecological consequences of urbanization.

**Graduate Researcher** **Fall 2018 – Summer 2023**
*The Earley Lab, University of Alabama*

- Quantifies the impact of abiotic variables on levels of gene flow and genetic divergence between populations of *Kryptolebias marmoratus*.
- Investigates eDNA concentration's relation to density as a proxy for species abundance.
- Evaluates the abiotic and biotic factors which impact occupancy and abundance in *Kryptolebias marmoratus* through ecological niche modeling and Bayesian hierarchal models.
- Identifies the genomic and epigenomic variants driving sexual plasticity while quantifying associated gene expression patterns.

**Undergraduate Researcher** **Fall 2016 - Summer 2018**
*Freshwater Ecology Lab, University of South Florida*

- Surveyed and identified littoral benthic invertebrates throughout the Tampa Bay area to assess urban pond health.
- Sampled, identified, and measured over 5,000 fish to assess the difference between urban pond types for direct application within urban planning.
- Assessed differences between pond types for both benthic invertebrate and littoral fish community composition and biodiversity.
- Completed geospatial analysis to derive pond characteristics from satellite imagery.

**Undergraduate Researcher** **Fall 2012 - Spring 2013**
*Saint Leo University, Saint Leo, FL*

- Quantified mangrove forest heath throughout Tampa Bay using quartile plots.
- Measured the abundance and distribution of galls located on red mangroves throughout Tampa Bay.
- Investigated the relationship between salinity and gall formation on red mangroves.
- Aided in artificial oyster reef experiments to quantify the abiotic factors affecting reef success.

**Research Volunteer** **Fall 2012 - Spring 2013**
*Mangrove Watch,*
*Tampa, FL*

- Conducted mangrove forest surveys throughout Tampa Bay.
- Employed video surveillance of shoreline habitat throughout Tampa Bay to assess mangrove health.
- Quantified mangrove forest health through individual tree tagging and measurements.
- Led groups of volunteers in collecting key data to serve as mangrove forest health baselines used in long-term monitoring programs.

## TEACHING EXPERIENCE

***University of Alabama***
*Instructor of Record* **Fall 2018 – Summer 2022**

- Biology II Laboratory
- Biology II Laboratory for Non-majors
- Introduction to Principles of Biology

*Instructor of Record indicates that the individual formatted, developed, and presented instructional material in compliance with department and course guidelines, while also being solely responsible for grading and course management.

*Graduate Teaching Assistant* **Spring 2021**

- Research Seminars in Biology
- Biological Sciences Seminar

## PROFESSIONAL MEMBERSHIPS

| | |
|---|---|
| Society for Molecular Biology & Evolution (SMBE) | **2022 - Present** |
| The Ecological Society of America (ESA) | **2022 - Present** |
| The International Biogeography Society | **2022 - Present** |
| The Society for Integrative and Comparative Biology (SICB) | **2022 - Present** |
| The Research Coordinate Network for Evolution in Changing Seas (RCN-ECS) | **2020 - Present** |
| The Society for the Study of Evolution (SSE) | **2020 - Present** |
| Sigma Xi | **2020 - Present** |
| The American Association for the Advancement of Science (AAAS) | **2018 – 2022** |

The Fisheries Society of the British Isles (FSBI) **2018 - Present**
University of Alabama Biology Graduate Student Association (UA BGSA) **2018 - 2023**

## PROFESSIONAL SERVICE

### *Internal*

UA Biological Sciences Diversity, Equity, & Inclusion Committee **Fall 2021 – Summer 2023**
Graduate Student Representative

### *External*

1000 Genomes ONT Sequencing Consortium **2022 - Present**
Member

RCN-ECS 2022 Reading Group - Temporal Data and Evolutionary Change **Spring 2022**
Co-Organizer

### *Journals Reviewed for:*

*Ecological Processes* (11/21)
*Molecular Phylogenetics and Evolution* (4/22)

### *Conferences Reviewed for:*

*Ecological Society of America – Portland (2023)*

## OUTREACH

Denver Metro Regional Science & Engineering Fair **Spring 2022**
*Volunteer Judge*

Meet a Bama Biologist **Spring 2021 – Summer 2023**
*Scientific Volunteer*

UA Undergraduate Research & Creative Activity Conference **Spring 2021, 2022**
*Volunteer Judge*

UA STEM Showcase **Spring 2021**
*Volunteer Judge*

Letters to a Pre-Scientist **Fall 2020 - Present**
*Scientific Volunteer*

UA SafeZone **Fall 2019 – Summer 2023**
*Ally/Trainer*

The University of Alabama's Night at the Museum **Spring 2019**
*Volunteer Biology Instructor*

Skype a Scientist **Spring 2018 - Present**
*Scientific Volunteer*

## REFERENCES

Dr. Ryan Earley
Associate Professor
The University of Alabama
Box 870344
300 Hackberry Lane
Tuscaloosa, AL 35487
(559) 451-6800
rlearley@ua.edu

Dr. William Ellis
Assistant Professor
University of South Florida
SCA 108
4202 East Fowler Avenue
Tampa, FL 33620
(727) 480-5472
wellis@usf.edu

Dr. Malin Pinsky
Associate Professor
Rutgers University
ENR 130
14 College Farm Rd
New Brunswick, NJ 08901
(848) 932-8242
malin.pinsky@rutgers.edu